Aaron Hong

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Education

University of Southern California

B.S. in Computer Science Games

August 2013 - May 2017

Skills

Languages: C#, Java, Kotlin, TypeScript, JavaScript

Engines & Graphics: Unity, Unreal Engine, Three.js, Cg

Web & Cloud: React, Node.js, AWS, HTML5 / CSS3

Hardware & 3D Design: Autodesk Fusion 360, Blender, EAGLE

2D & Technical Art: Figma, Affinity Designer, Lottie

Tools: Git, JIRA

Interests

PC Gaming

Jazz Bass

Mechanical Keyboards

Ping-Pong

VOCALOID Music

Work Experience

Amazon - Software Development Engineer II (Android) **May 2020 - July 2025**

- **Led end-to-end technical design** for Amazon Kids features, defining full-stack architectural requirements and negotiating API contracts with backend teams to ensure scalable implementation.
- **Directed incident response for high-severity production issues**, utilizing telemetry to isolate cross-stack failures and coordinating with backend service owners to expedite resolution.
- **Expanded engineering scope beyond mobile** by contributing directly to backend AWS microservices via "away team" initiatives, deploying server-side logic to unblock critical path features.
- **Modernized legacy codebases** by refactoring UI components to Jetpack Compose, significantly improving rendering performance and developer velocity.
- **Maintained high code quality** by actively contributing to code reviews (PRs), collaborating with the team to identify potential regressions and ensure consistent design patterns.

Pinscreen - Software Engineer (Front-End / Application) **February 2018 - January 2020**

- **Led development of the flagship iOS demo application** in Unity, translating proprietary computer vision research into a consumer-facing product capable of generating photorealistic 3D avatars from single-image selfies.
  Before (Early 2018) After Redesign (Mid 2018) Final App (2019)
- **Engineered custom mesh processing algorithms** in Unity and UE4, writing a raw .obj parser with logic for topology remapping, polycount reduction, and glTF serialization to optimize high-fidelity assets.
- **Built a runtime character assembly system**, developing the logic to programmatically stitch and weld AI-generated head meshes onto standardized body assets to ensure seamless animation.
- **Integrated proprietary face-tracking algorithms** via native C++ plugins (DLLs), mapping raw data streams to avatar blendshapes to drive real-time animation without relying on platform-specific APIs.
- **Developed cross-platform 3D proofs-of-concept**, including a hyper-realistic desktop demo in Unreal Engine 4, and a React.js web application that generated and rendered GLTF avatars via Three.js.
- **Redesigned and deployed the company website** in React, collaborating with artists to translate UI/UX designs into a responsive frontend.
  Before (Early 2018) First Redesign (Mid 2018) Final Website (2019)

Mild Beast Games - Co-Founder and Game Developer **June 2015 - February 2018**

- **Shipped the BAFTA-winning title *AT SUNDOWN: Shots in the Dark*** on Steam, PS4, Xbox One, and Switch, driving the entire development lifecycle and managing technical compliance with publishing partners.
- **Implemented the core stealth mechanic**, writing custom dynamic shaders to determine visibility for the game's core stealth mechanic.
- **Programmed combat systems**, implementing a diverse arsenal of weapons and power-ups with unique behaviors.
- **Created custom tech art systems**, including rope rendering and screen-space fluid simulation, to establish the game's unique aesthetic identity.
- **Developed a proprietary Level Editor**, enabling rapid iteration on map layouts and gameplay scenarios to accelerate the level design process.
- **Engineered the enemy AI system**, implementing state-machine behaviors and pathfinding for CPU-controlled opponents.
- **Showcased the game** at various industry conferences and events.

Other Projects

Synth Labs - Founder **March 2022 - Current**

- **Founded and scaled a consumer electronics business**, designing and shipping over 1500 units of custom peripherals, including a keyboard design, macropad, and original keycap profile, to a global customer base.
- **Engineered custom hardware schematics**, utilizing Fusion 360 for CAD modeling and EAGLE for PCB design to manage the full production lifecycle from prototype to final assembly.
- **Developed embedded firmware** in C using QMK/VIA, programming custom rotary encoder behaviors, dynamic keymaps, and RGB lighting modes to enhance user customization.
- **Directed creative strategy**, producing photorealistic 3D assets in Blender, integrating a 3D product configurator for the macropad, and managing design collaborations for the SLK Dessau keycap set.
- **Managed end-to-end operations**, coordinating with manufacturers for CNC, injection molding, and soft goods (custom carrying cases) while executing quality control.

Ongoing Projects